



ITI Cmos Analog IC Design

Lab 05

Simple vs Wide Swing (Low Compliance) Cascode Current Mirror

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Section 7 online

Part 1: Sizing Tradeoffs Using SA

1.1 Design Specifications

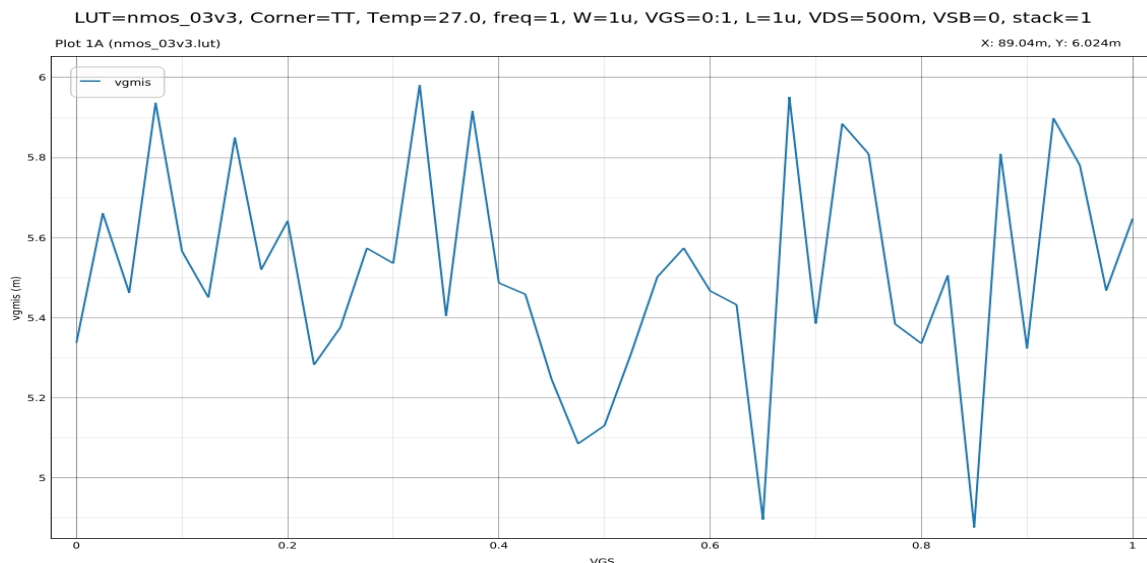
Parameter	
Current direction (source/sink)	Sink
Input Current	$10\mu A$
Output Current	$20\mu A$
% Change in Current for $\Delta V_{out} = 1V$	$< 10\%$
Percent mismatch: $\sigma(I_{out})/I_{out}$	$\leq 2\%$
Compliance voltage	$\leq 150mV$
Area	Minimize

- Sinking current mirror means is **Nmos type**

1.2 Required λ Calculation

$$I_D = I_{Dsat} (1 + \lambda \Delta V_{DS})$$
$$\frac{\Delta I_{out}}{I_{out}} = \lambda \Delta V_{DS}$$
$$\Delta V_{DS} = 1V, \frac{\Delta I_{out}}{I_{out}} < 10\%$$
$$\lambda \cdot (1V) < 0.10 \Rightarrow \lambda < 0.1 V^{-1}$$

1.3 σV_T and v_{gmis} vs VGS Plot



- Average $v_{g\text{ mis}}$ from plot is **5.45 mV**

- σ_{VT} Model File

```

46931 .subckt nfet_03v3 d g s b w=1e-5 l=2.8e-7 nf=1 m=1 as=0 ad=0 ps=0 pd=0 nrd=0 nrs=0
      par=1 dtemp=0 sa=0 sb=0 sd=0
46932 .param par_vth=0.007148
46933 .param par_k=0.007008
46934 .param par_l=1.5e-7
46935 .param par_w=-1e-7
46936 .param par_leff='l-par_l'
46937 .param par_weff='par*(w-par_w)'
46938 .param p_sqrtarea='sqrt((par_leff)*(par_weff))'
46939 .param var_k='0.7071*par_k*1e-06/p_sqrtarea'
46940 .param mis_k='agauss(0,var_k,1)'
46941 .param var_vth='0.7071*par_vth*1e-06/p_sqrtarea'
46942 .param mis_vth='agauss(0,var_vth,1)'
46943 m0 d g s b nfet_03v3 ad=ad as=as delvto='mis_vth*sw_stat_mismatch' l=l m=m
      mulu0='1-mis_k*sw_stat_mismatch' nf=nf nrd=nrd nrs=nrs pd=pd ps=ps sa=sa sb=sb
      sd=sd w=w
46944 .ends nfet_03v3
46945 .subckt pfet_03v3 d g s b w=1e-5 l=2.8e-7 nf=1 m=1 as=0 ad=0 ps=0 pd=0 nrd=0 nrs=0
      par=1 dtemp=0 sa=0 sb=0 sd=0
46946 .param par_vth=0.00666
46947 .param par_k=0.002833
46948 .param par_l=1.5e-7
46949 .param par_w=-1e-7
46950 .param par_leff='l-par_l'

```

$$\sigma_{VT}(\text{var_vth}) = \frac{0.7071 * 0.007148 * 10^{-6}}{\sqrt{1u * 1u}} = \mathbf{5.289 \text{ mV}}$$

- ✧ *the $V_{G\text{mis}}$ from ADT is larger than the one from model file , because ADT monte carlo simulation can include different mismatch sources not only vth.*

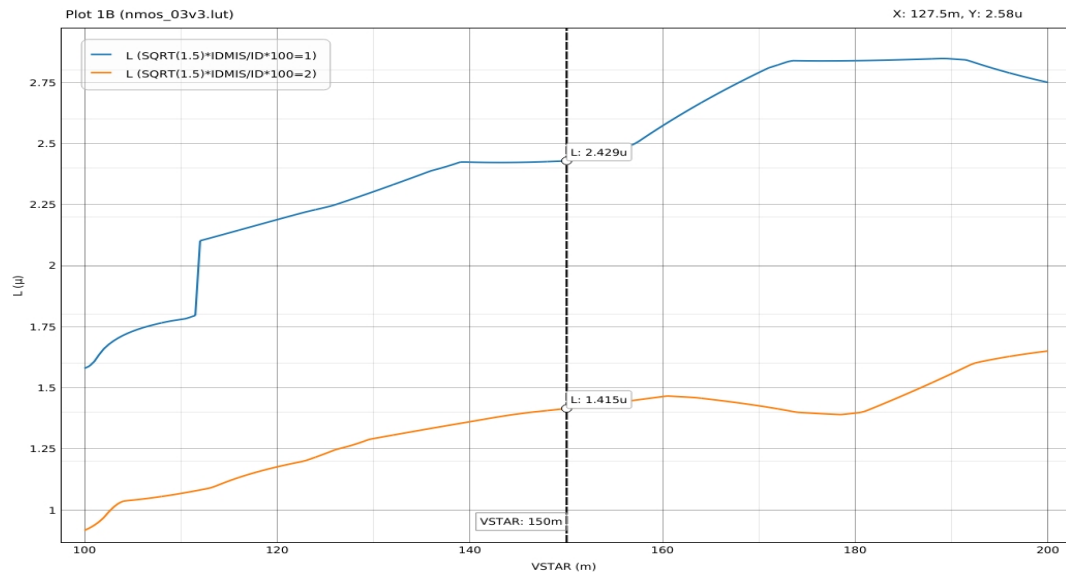
$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2}$$

1.4 Sizing Trade-off Plots (L, W, Area, λ)

ID	10u
Vstar	100m:200m
$\sqrt{1.5} * id_{\text{mis}} / ID * 100$	1, 2
VDS	0.5
VSB	0
Stack	1

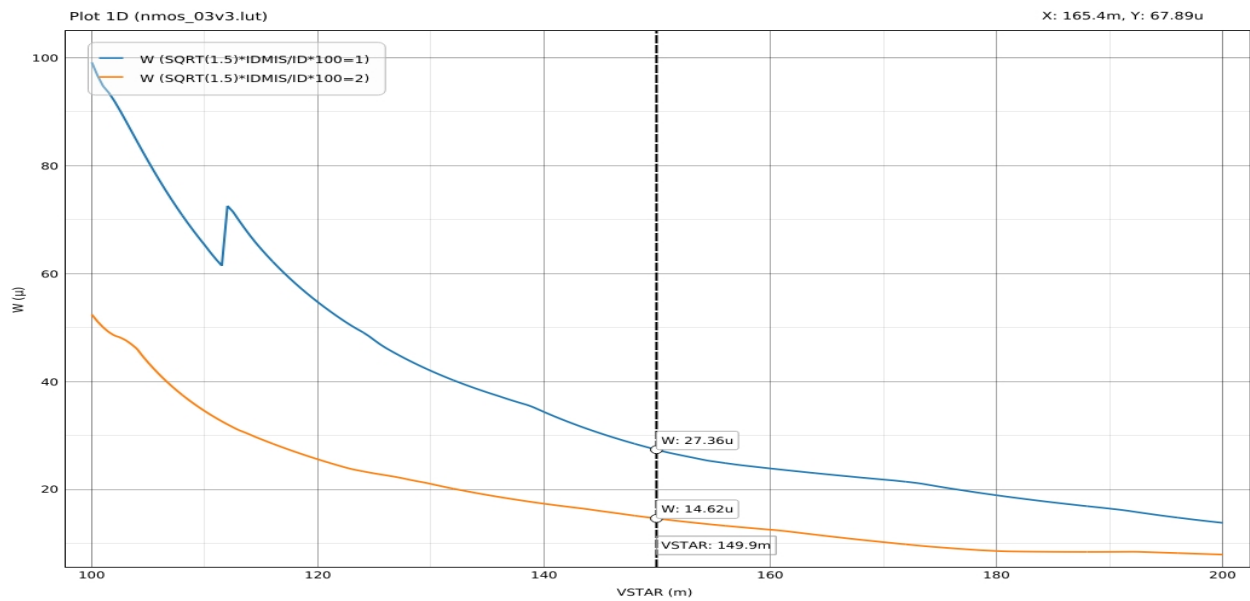
● L VS V*

, Corner=TT, Temp=27.0, freq=1, ID=10u, VSTAR=100m:200m, SQRT(1.5)*IDMIS/ID*100=1/2, VDS=500m,



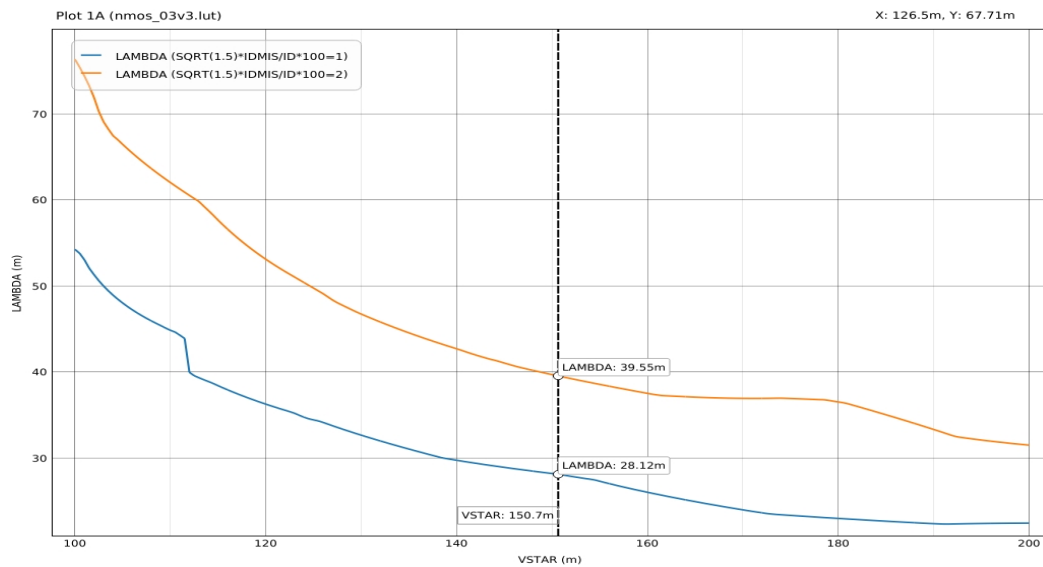
● W VS V*

, Corner=TT, Temp=27.0, freq=1, ID=10u, VSTAR=100m:200m, SQRT(1.5)*IDMIS/ID*100=1/2, VDS=500m,



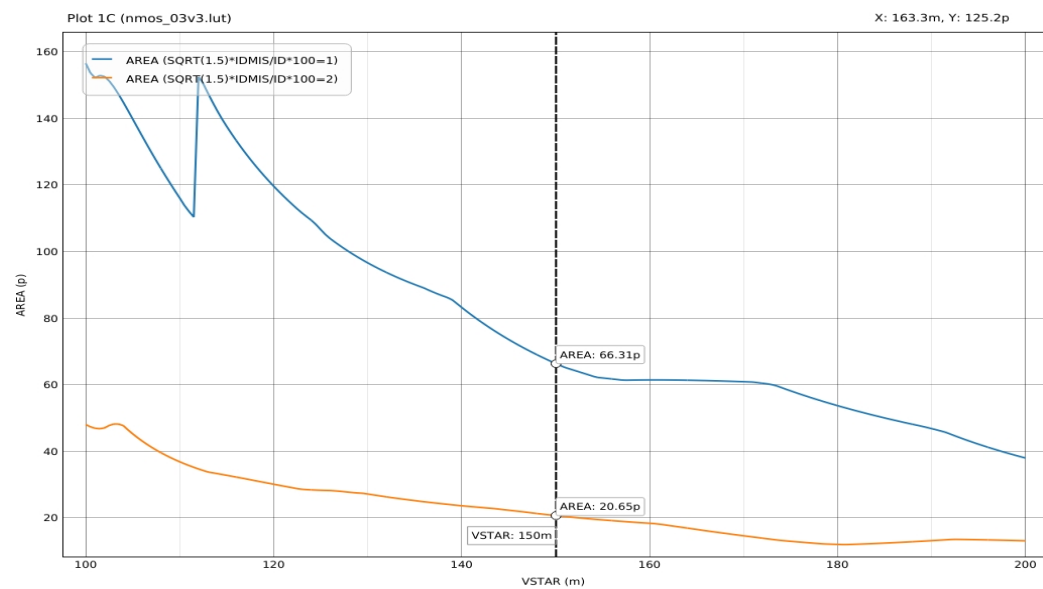
● λ VS V^*

, Corner=TT, Temp=27.0, freq=1, ID=10u, VSTAR=100m:200m, SQRT(1.5)*IDMIS/ID*100=1/2, VDS=500m,



● Area VS V^*

, Corner=TT, Temp=27.0, freq=1, ID=10u, VSTAR=100m:200m, SQRT(1.5)*IDMIS/ID*100=1/2, VDS=500m,



✧ From previous plots indicates that at highest V^* is good for mismatch, since λ is decreasing and that is good for mismatch ($\lambda = 1 / V_a$) and λ is 39.5 m so it satisfies the requirements, area increasing with V^* and that because square law at constant drain current when V^* goes up so W/L must go down but W decreasing more than increasing L so total area is decreased

✧ When trying satisfying the lambda at 0.1

$\sigma(I_{out})/I_{out}$ will equal to 5.526

Which not Meets requirements

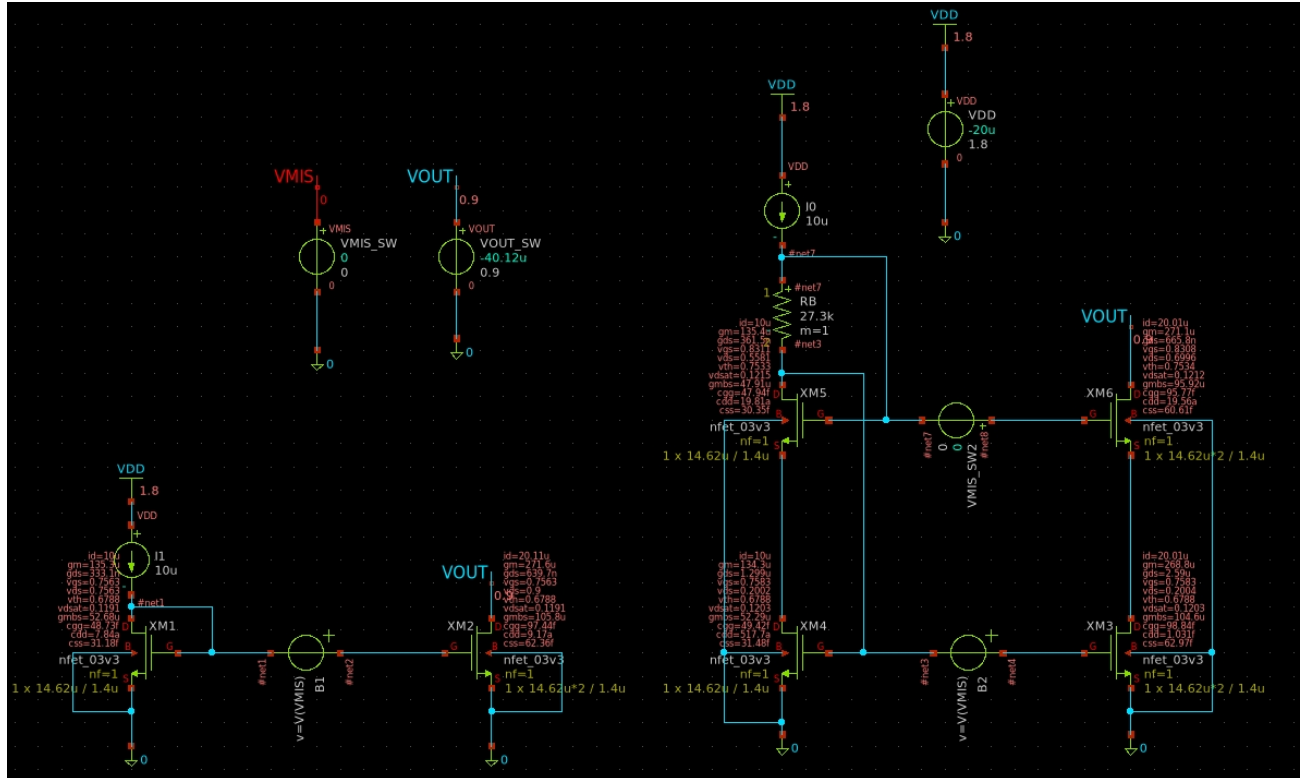
ID	10u	?
Vstar	150m	?
LAMBDA	0.1	?
VDS	0.5	?
VSB	0	?
Stack	1	?

✧ So From graphs at $\sigma(I_{out})/I_{out} = 2$,

$W = 14.62u$, $L = 1.4u$, $\lambda = 39.5m$

Part 2: Current Mirror Simulation

2.1 Schematic Construction

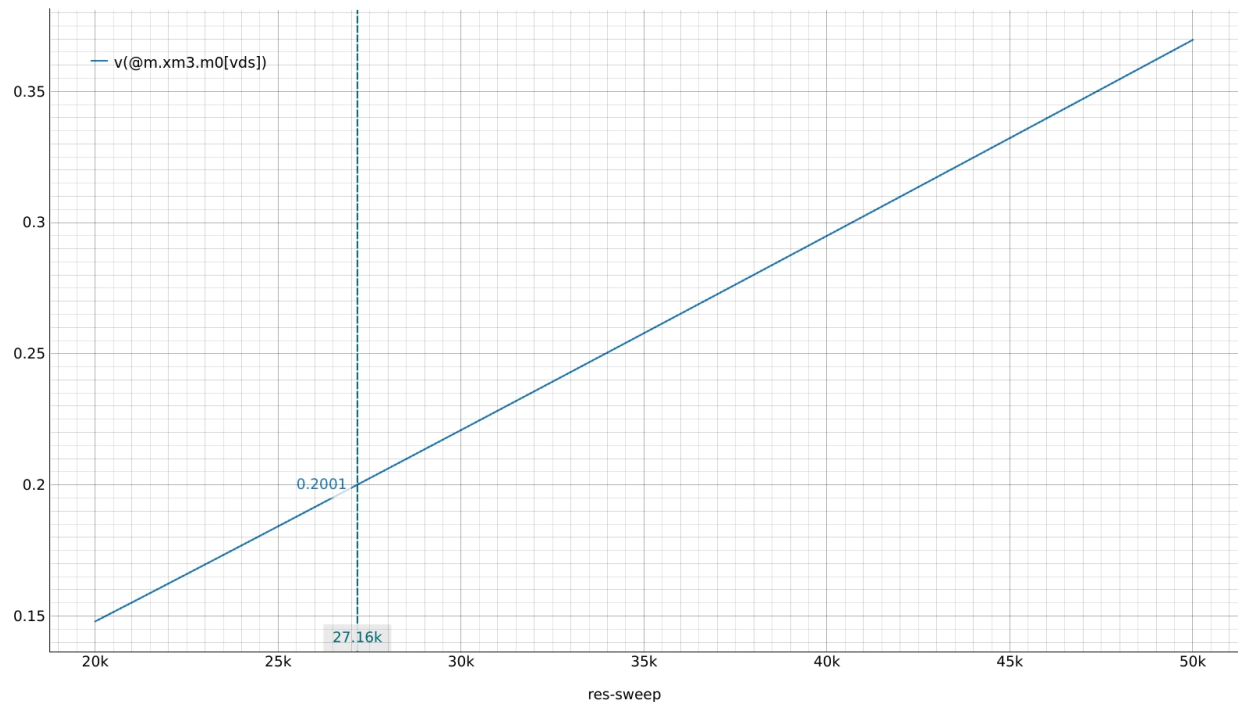


2.2 Design and Operating Point (OP) Analysis

2.2.1 RB Hand Calculation

$$RB = \frac{v_{gs5} + v_{ds2} - v_{gs2}}{i_b} = \frac{v_{ds2}}{i_b} = \frac{200m}{10u} = 20k$$

2.2.2 DC Sweep – VDS3 vs RB



- ✧ From graph R_B is 27.16 k
- ✧ Value of R_B in hand analysis is less because we assume $v_{gs5} = v_{gs2}$, however v_{gs5} is more than v_{gs2} because it suffers from body effect

2.2.3 Operating Point

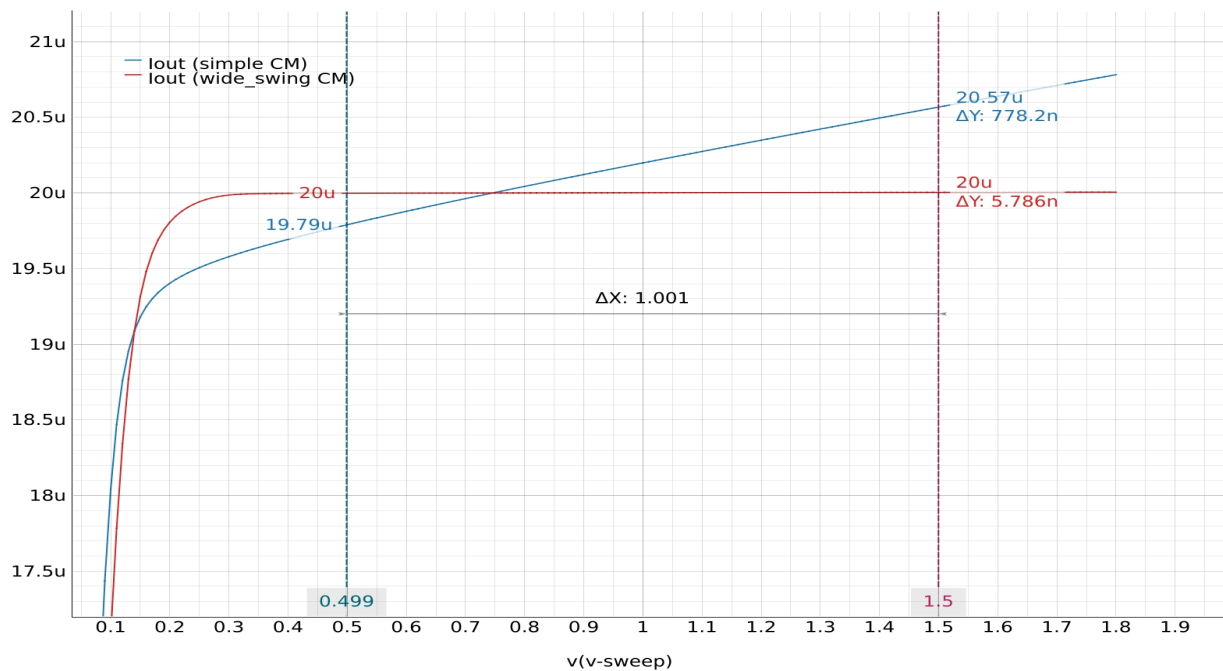
```
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.xm6.m0      m.xm5.m0      m.xm4.m0
model       nfet_03v3.14   nfet_03v3.14   nfet_03v3.14
id          2.00143e-05  1e-05         1e-05
vgs         0.830823  0.831087      0.758269
vds         0.699552  0.558087      0.200179
vth         0.753411  0.753317      0.678805
vdsat       0.121234  0.121462      0.12029
gm          0.000271087  0.000135408   0.00013429
gds         6.6581e-07  3.61482e-07   1.29947e-06
gmbs        9.59249e-05  4.79136e-05   5.22856e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.xm3.m0      m.xm2.m0      m.xm1.m0
model       nfet_03v3.14   nfet_03v3.14   nfet_03v3.14
id          2.00143e-05  2.01074e-05    1e-05
vgs         0.758269  0.756326      0.756326
vds         0.200444  0.899998      0.756325
vth         0.678805  0.678805      0.678805
vdsat       0.12029  0.11906       0.11906
gm          0.000268779  0.000271616   0.000135271
gds         2.5899e-06  6.3966e-07    3.33087e-07
gmbs        0.000104648  0.000105793   5.26818e-05
```

- ✧ From operating points above all transistors operate in saturation.

2.3 DC Sweep – Iout vs VOUT

2.3.1 Iout vs VOUT Plot and Compliance Voltage



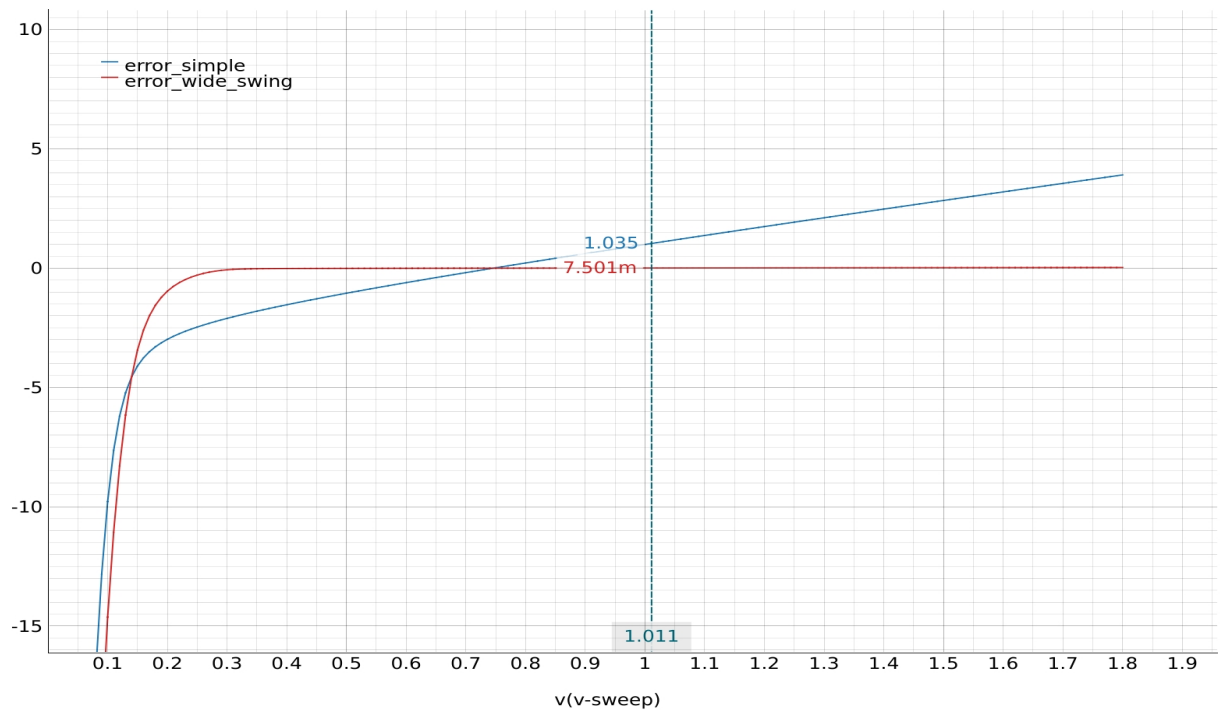
- ✧ The Simple current mirror increase with V_{out} , but the wide swing current almost constant with V_{out} , and this difference because of difference of R_{out} of two circuit.
- ✧ The compliance voltage of simple CM is less which equal approximately 170m, the compliance voltage of wide-swing approximately equal to 200m.
- ✧ I_{out} of the simple CM is exactly equal to $I_B \cdot 2$ at $v_{ds} = 740m$, which equal to V_{GS} so the two transistor of current mirror is exactly same (v_{gs}, v_{ds})

2.3.2 Percent Change in Iout (VOUT = 0.5 V to 1.5 V)

```
No. of Data Rows : 181
Simple Current Mirror Simulation Results
i1          = 1.98319e-05
i2          = 2.04731e-05
perc_change = 3.181642e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.00121e-05
i2          = 2.00169e-05
perc_change = 2.398262e-02
```

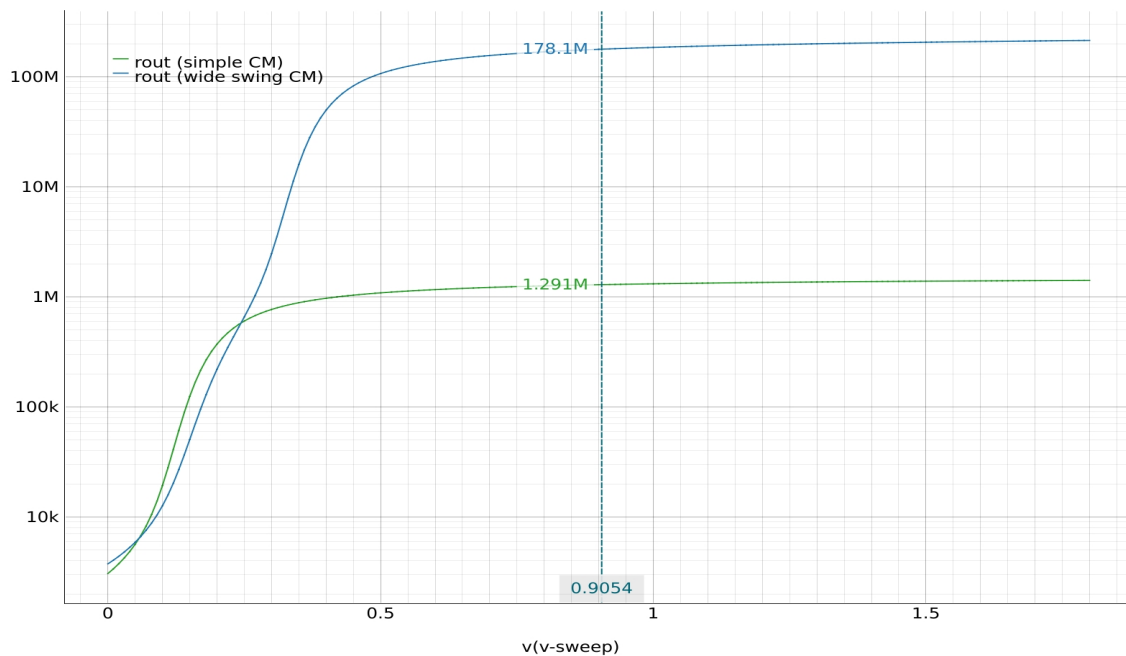
- ✧ The simple current mirror get **3.1816%** change in output current, in part 1 we put it **2%**, and this due to r_{clm} and output resistance.
- ✧ Wide swing current mirror give a much more better current change than simple one, but from tradeoff point WS need more compliance voltage (headroom)

2.3.3 Percent Error in I_{out} vs V_{OUT}



✧ As expected the percent error in simple current mirror is higher than wide swing which approximately equal zero.

2.3.4 R_{out} vs V_{OUT} Plot



- ✧ At $v_{out} = .9$, $R_{out} \text{ (Simple cm)} = 1.291 \text{ M}\Omega$
 $R_{out} \text{ (WS cm)} = 178.1 \text{ M}\Omega$
- ✧ Output resistance of cascode is much higher than CM (xx 138)
- ✧ Does R_{out} change with V_{OUT} ? Why? R_{out} change with V_{out} , at low v_{out} transistor are not in saturation so output resistance is small, at high v_{out} all transistor operate in saturation and also deep in saturation which give a high output resistance.

✧ **Analytical R_{out} Calculation**
using parameter from op analysis:

$$R_{out}(CM) = r_o = 1 / g_{ds} = 1.293 \text{ M}\Omega$$

$$R_{out}(WS) = g_{m6}(r_{o3} * r_{o6}) = 157.66 \text{ M}\Omega$$

	Analyticaly	simulations
Simple current mirror	1.293 MΩ	1.291 MΩ
Wide-swing current mirror	157.66 MΩ	178.1 MΩ

2.4 Mismatch Analysis

2.4.1 VMIS_SW Sweep (current mirror input)

V_{mis} from 0 to $\sqrt{1.5} * 1.24\text{m}$ (v_{gmis}) = 1.5m

// v_{gmis} taken from ADT

- Simulation

```
no. of data rows = 10
Simple Current Mirror Simulation Results
i1          = 2.01074e-05 at= 0.00000e+00
i2          = 2.05176e-05 at= 1.50000e-03
perc_change = 2.019645e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.00143e-05 at= 0.00000e+00
i2          = 2.04173e-05 at= 1.50000e-03
perc_change = 1.993240e+00
binary raw file "mismatch.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

- Analytical Calculation

using parameter in op analysis:

for simple cm

$$\Delta I_{out} = \frac{G_m * V_{MIS}}{I_{OUT}} = \frac{g_{m2} * 1.5\text{m}}{20 \mu} = 2.037 \%$$

For wide swing cm

$$\Delta I_{out} = \frac{G_m * V_{MIS}}{I_{OUT}} = \frac{g_{m3} * 1.5m}{20 \mu} = 2.015 \%$$

- comparison

	Analytically mismatch	Simulations mismatch
Simple current mirror	2.037 %	2.0169 %
Wide-swing current mirror	2.016 %	1.9932 %

2.4.2 VMIS_SW2 Sweep (cascode mirror input)

Vmis from 0 to sqrt(1.5)*1.24m (vgmis) = 1.5m

// vgmis taken from ADT

- Simulation

```
NO. OF DATA ROWS : 10
Wide Swing Current Mirror Simulation Results (Cascode Mismatch)
i1          = 2.00143e-05 at= 0.00000e+00
i2          = 2.00175e-05 at= 1.70000e-03
perc_change = 1.593729e-02
```

- Analytical Calculation

using parameter in op analysis:

for cascode wide swing :

$$\frac{\Delta I_{OUT}}{I_{OUT}} = \frac{G_m * V_{MIS}}{I_{OUT}} = \frac{\frac{g_{m6}}{1 + g_{m6} r_{o3}} * 1.5m}{20 \mu} = 1.9 * 10^{-2} \%$$

- comparison

	Analytically mismatch	Simulations mismatch
Wide-swing Cascode	19.2m %	15.93m %

✧ Which mismatch contribution is more pronounced? Why?

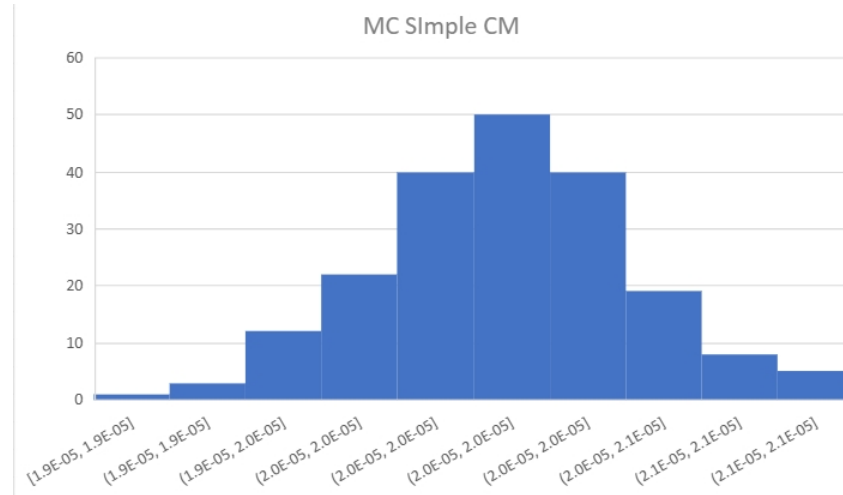
The current mirror mismatch is more larger because the variation of voltage is multiply in gm which a large value , otherwise in cascode the voltage multiply in degenerated GM which is very low.

✧ Which design decision is better: setting the same W and L for the mirror and cascode devices? Or using larger W and L for the current mirror devices? second option is very good design decision , because increasing area will reduce the mismatch, which is good for current mirror, because it has a big mismatch than cascode devices.

2.5 Monte Carlo (MC) Simulation

- For Simple Current mirror

- Histogram of I_{out}



- standard deviation of current

	A	B	C	D
1	1.97E-05	mean	std	std_%
2	2.04E-05	2.01E-05	3.69045E-07	1.84E+00
3	2.02E-05			
4	2.03E-05			
5	2.01E-05			
6	1.97E-05			
7	2.07E-05			
8	1.99E-05			
9	2.04E-05			
10	2.00E-05			

- analytical analysis

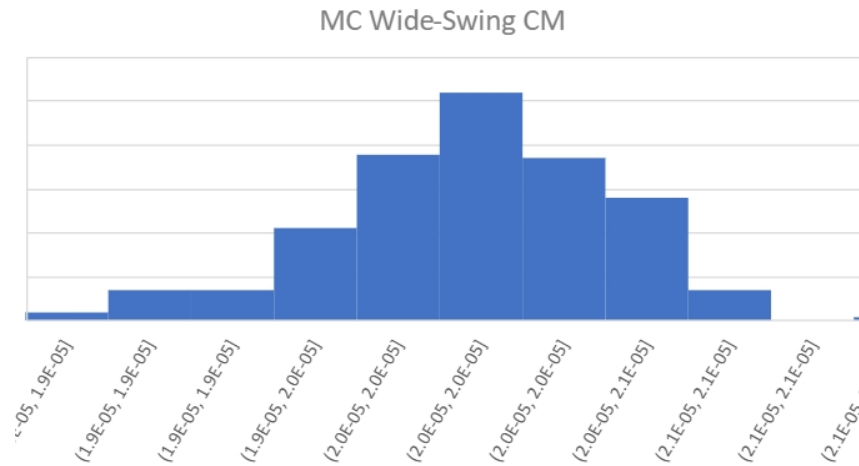
$$\frac{\sigma(I_{out})}{I_{out}} = \frac{\sqrt{1 + \frac{1}{m} * idmis}}{ID}$$

$$\frac{\sigma(I_{out})}{I_{out}} = \frac{\sqrt{1 + \frac{1}{m} * gm * \sigma(V_{TH})}}{ID} = \frac{\sqrt{2} * \frac{2id}{V^*} * \sigma(V_{TH})}{ID} = \frac{\sqrt{1.5} * 2 * \sigma(V_{TH})}{V^*}$$

$$\frac{\sqrt{1.5} * 2 * \sigma(V_{TH})}{V^*} = \frac{\sqrt{1.5} * 2 * 1.24m}{150 m} = 2.021 \%$$

- For wide swig Current mirror

- Histogram of I_{out}



- standard deviation of current

	A	B	C	D
1	2.01E-05	mean	std	std_ %
2	1.96E-05	2.00E-05	4.21E-07	2.10E+00
3	2.02E-05			
4	1.99E-05			
5	1.97E-05			
6	2.00E-05			
7	1.92E-05			
8	1.99E-05			
9	2.07E-05			
10	2.01E-05			

- expected analytical analysis

$$\frac{\sigma(I_{out})}{I_{out}} = \frac{\sqrt{1 + \frac{1}{m} * idmis}}{ID}$$

$$\frac{\sigma(I_{out})}{I_{out}} = \frac{\sqrt{1 + \frac{1}{m} * gm * \sigma(VTH)}}{ID} = \frac{\sqrt{2} * \frac{2id}{V^*} * \sigma(VTH)}{ID} = \frac{\sqrt{1.5} * 2 * \sigma(VTH)}{V^*}$$

$$\frac{\sqrt{1.5} * 2 * \sigma(VTH)}{V^*} = \frac{\sqrt{1.5} * 2 * 1.24m}{150 m} = 2 \%$$

	<i>Simple current mirror</i>		<i>Wide swing current mirror</i>	
	Simulated	analyticaly	Simulated	analyticaly
$\frac{\sigma(I_{out})}{I_{out}}$	1.84 %	2.021 %	2.1 %	2 %